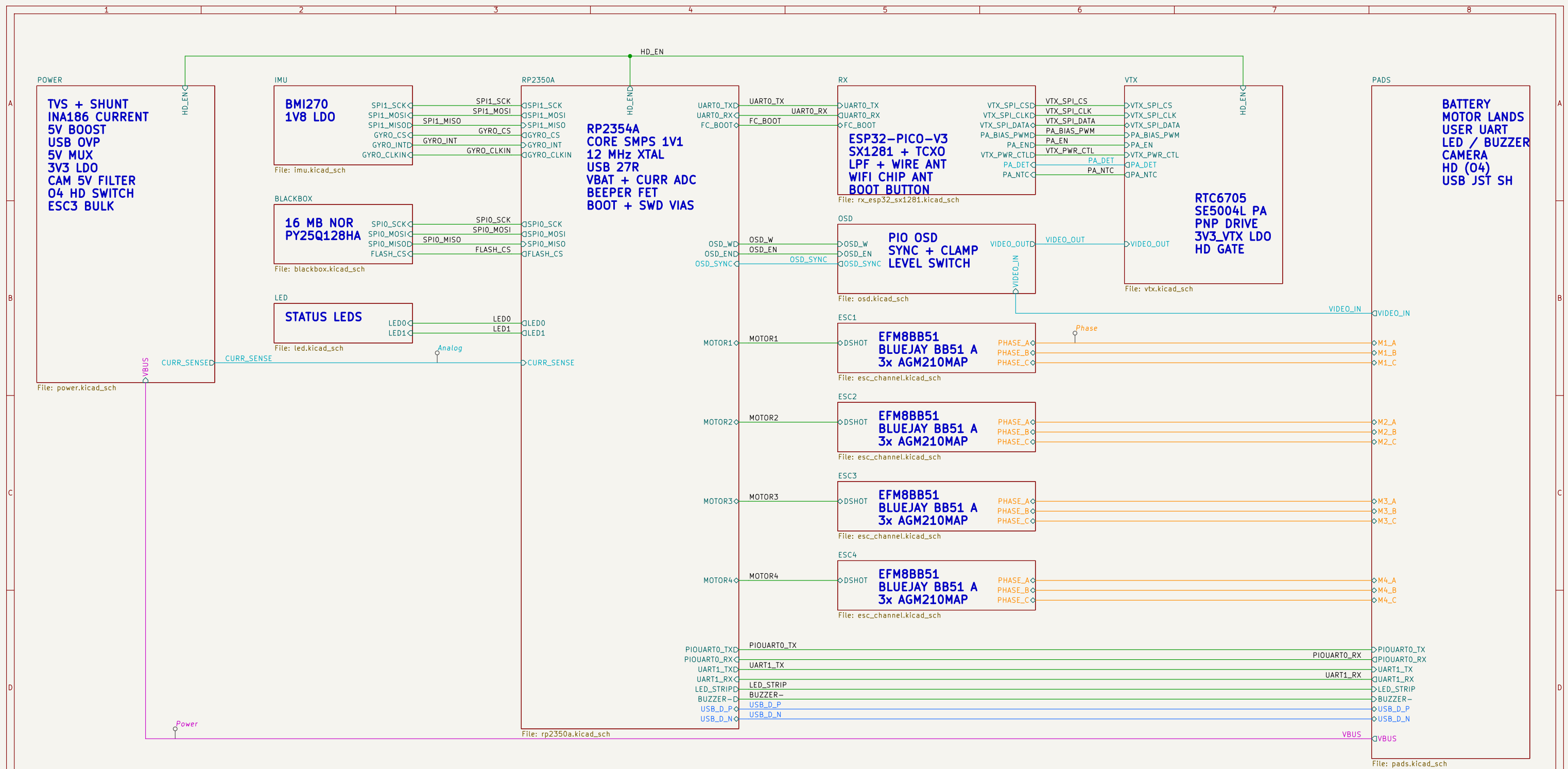




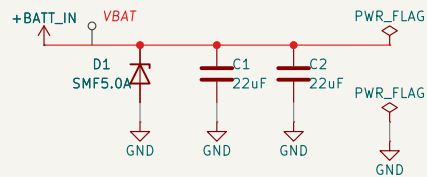
POWER TABLE

Rail	Source	Nominal	Budget (spec 5)	Feeds
+BATT_IN	battery pads J2 / J3 (15)	3.0–4.35 V	4x ESC + 2.0 A board	TVS, pad bulk, shunt R1
+BATT	shunt R1 0.5 mOhm	cell	ESC + boost in 1.4–3.0 A	ESC1–4, boost, O4 EN, VBAT ADC, PA_VCC
/VBUS	USB J31 pin 4 (hier. net)	5 V	0.42 A, USB only	U24 OVP
+5V_USB	U24 SGM40661 OVP	5 V	as VBUS	U3 VIN1, PR1 divider
+5V_BST	U2 TPS61022 boost	5.15 V	1.82 A at 3.0 V in	U3 VIN2, U5 O4 switch
+5V	U3 TPS2116 mux	5 V	0.70 A analog, 1.48 A HD	3V3 LDOs, CAM filter, 5V pads
+5V_CAM	FB1 ferrite from +5V	5 V	0.12 A	camera pad + plug
+5V_HD	U5 TPS22810 (HD_EN, cell)	5 V	1.20 A O4 Lite	VHD pad
+3V3	U4 LP5912–3.3	3.3 V	0.20 A (0.48 A Wi-Fi bench)	FC, IMU, NOR, OSD, RX, LEDs, INA186
+3V3_VTX	U21 LP5912–3.3 (off in HD)	3.3 V	0.10 A	RTC6705, PA drive, PA ref
+1V8	U12 TPS7A2018	1.8 V	< 1 mA	BMI270
+1V1	RP2354A core SMPS	1.1 V	internal	RP2354A DVDD
PA_VCC	R67 selector, +BATT default	cell	0.65 A at 27 dBm	SE5004L PA (local, VTX)
GND	battery –	0 V	–	all sheets

Generated from hardware/sch_contract.json by hardware/tools/sch_root.py: do not edit by hand.
 Root wires carry the project-wide net names (root label = net /NAME). Motor phases are named by
 the ESC sheets (/ESCn/PHASE_x). Rails are power symbols (table); VBUS is a hierarchical net.
 ESC1~ESC4 drive M1 rear right, M2 front right, M3 rear left, M4 front left (Betaflight quad-X); FC pin
 name: research/PINAD.md.

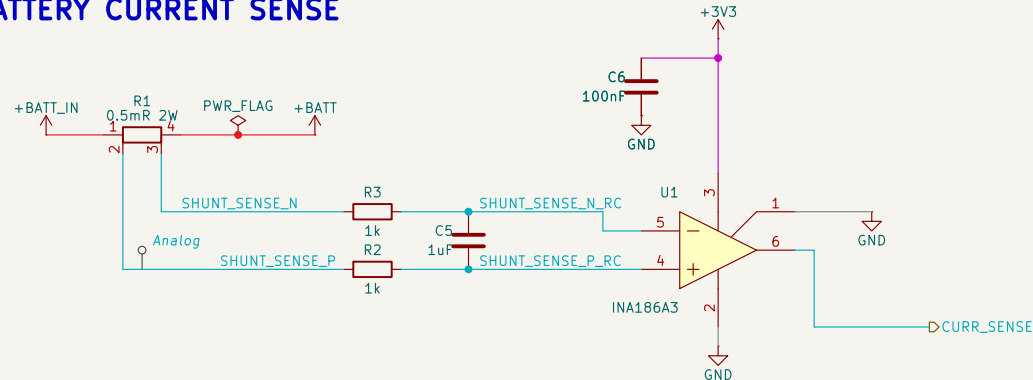
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BATTERY INPUT AND TVS



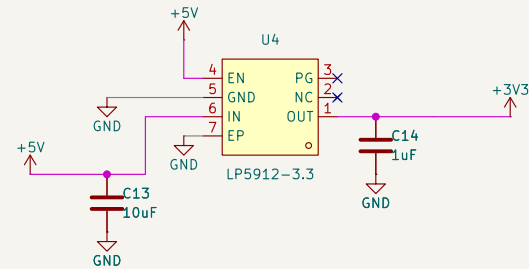
Source: OpenAI/O1/hardware/OpenAI_kicad_sch (battery entry), adapted.
+BATT_IN = battery pads J2 (+) / -/- on the PADS sheet, 1S LiPo/LiHV
3.0-4.35 V; PWR_FLAG here (the pads are passive). No reverse-polarity FET (spec 4.1) – user-fitted pigtail, risk in README.
D1 SMF5.0A: VBR 6.40~7.07 V, VC 9.2 V at 21.7 A, 200 W 10/100ms us: takes one meter's regen energy (-0.2J/kg per rated pulse) on a battery disconnect; fallback 100W 10/100ms us: 0.2J/kg, 1V3C, 5% excess heat. No hold time, no 5.5-7.1V parts (EFMB VDD, TP561022, TPS2116 in rating: Matrix-parity risk, spec 15.4, C1/C2 22 uF 16 V XSR 0603 at the pads: 9–13 uF effective each at 4.35 V, Hot plug of a 4.35 V pack can overshoot to 5.0-5.4 V (V3c); damping fallback ~100 uF polymer at the pads: PE-8 waits for V3c.

BATTERY CURRENT SENSE



Source: OpenAIO/hardware/OpenAIO.kicad_sch (shunt + INA186A3, root sheet); input RC moved to 0201 and its two common-mode caps deleted (DS2C BR-07); the output RC (1k/100n) sits at the ADC pin (RP2350A sheet, R47/C60).
P4-9: R1 HC5206 0.5 mOhm on the four-pad Kelvin net-tie land (pad groups 1-2 and 3-4). SHUNT_SENSE_PC = pad 2 (battery side).
SHUNT_SENSE_N = pad 3: each holds only its inner sense pad and R2/R3; SHUNT_SENSE_PC/N_RC hold R2/R3, C5 and U1 (netlist checked); all four match the Analog and Kelvin DRU patterns. Route the pair on L1 or L6, never in2/13 (DRU). class Analog. Everything that draws current taps the R4.
R2/R3 1k + C5 1 uF differential: $c = 1 / (2 \pi \times f \times 20 \text{ kOhm} \times 1 \text{ uF}) = 80 \text{ Hz}$, $\text{ILB} < c = 3 \text{ nA} \times 1 \text{ kOhm} = 3 \text{ uV}$ (6 mA), $\text{VOS} < 50 \text{ uV}$ (0.1 A).
U1 INA186A3, gain 100: 50 mOhm $\times 100 = 50 \text{ mV/A}$, Betaflight ibata_scale 500: output to VS - 40 mV gives about 65 A full scale on +3V3; VCM -0.2..0.4 V covers the cell; REF to GND (unidirectional); C6 100 nF at VS. Shunt loss 0.2 W at 20 A. Scale checked on every prototype (V-pin).

3.3 V LDO FOR FC + RX



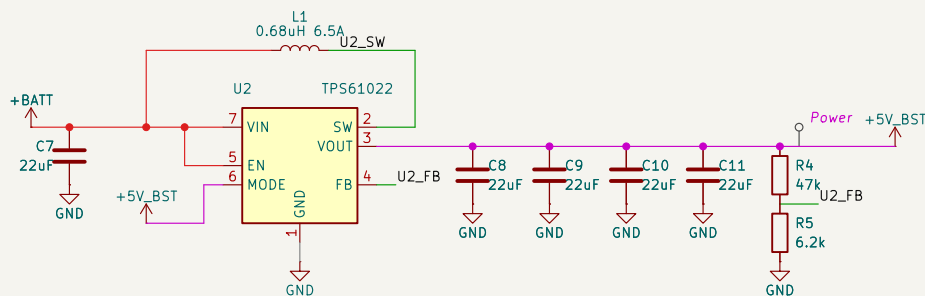
Source: OpenFC-Lite-Mini/hardware/power/kicad.sch (3.3V 0.5A MAIN LDO", LP5912-3.3), with the OpenDrone LP5912-3.3DRVR symbol (pins in package order: IN/EN on the left here, EN tied to +5V, no EN shared with another rail; U21 on VTX has its own EN, PG and NC open; EP to GND for heat (T1: GND or open)).

P4-17 / BR-14: C13 10 uF X6S 0402 is the one CIN of U4 and U21 (VTX sheet): place it between the two IN pins (-2.4 mm apart). 2.0 uF effective at 1.5 V (Murata curve) > the 0.7 uF LP5912 minimum.

C20 12-3V3 0.01 uF at 3.3 V (C82 10 uF 5 V, Murata CWS5C80J106M11 DC-bias curve). C58 4.7 uF 0402 X5R -2.0 V, 3x 1 uF 0201 X5R (C14, C61, C92) 1 uF each, 150 uF of X6S 0201 -0.085 uF each (last three typical, i); 8.7 uF typical, 7.1 uF at -2.0 V, 10.3 uF only with every part at +20 % -> Inside T1: 0.7-10 uF (SNSV470 7.6) but for that +3 % corner, and most of it is remote (5-30 mm of trace), which the LP5912 allows. Lever if a V-bench load step rings: PS7-06 (delete C92). +3V3_VTX is summed on the VTX sheet.

Load 0.20 A: (5.15 - 3.3) V x 0.2 A = 0.37 W, +26 K (RthJA 71.2 K/W). U4 UOT (project-lib LP5912 copy, power=not pin) drives +3V3: no PWR_FLAG.

5 V BOOST TPS61022



New block from TI SLVSDX7D (n sibling uses the TPS61022): look matched to the OpenFET-Lite–Mini buck blocks.

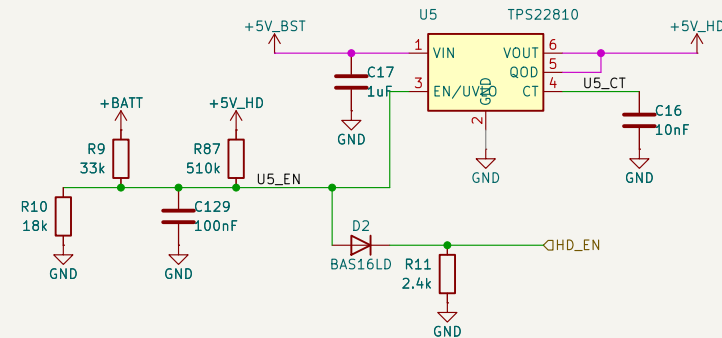
P4-7: VOUT = 0.600 V x (1 + 47k / 6.2k) = 1.548 V, with VFB 585–615 mV and 1 % resistors 4.93–5.37 V. R5 6.2k < 300 kOhm (TI). EN tied to VIN (+BATT) and nothing else (BR–11): same –0.3–7 V abs max, VEN_H <= 1.2 V, on whenever the cell is above UVLO. MODE = VOUT (+5V_BST): forced PWM, VMODE_H <= 1.2 V valid once VOUT > 2.2 V (start-up pass-through holds VOUT near VIN).

C7 22 uF CIN at VIN/GND. C8–C11 4x 22 uF X6S 0603: –20–28 uF effective at 1.5 V, inside TI's 10–50 uF and >= 20 uF at above 1.5 A (D15); no feed–forward cap (Cout 4 uF, VIN = 2.8 V, TI 8.2.2.4).

Min 6.5–10 A: 1.67/1.82/1.98 A out at 2.8/3.0/3.2 V (spec 4.2). Passes VIN to VOUT when VOUT (surge path, spec 4.1).

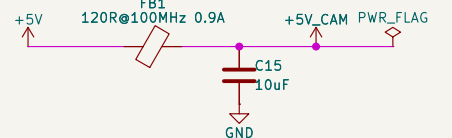
Layout (TI 10.1): Cout VOUT/GND pads at pins 3/1 (hot loop SW–VOUT–Cout–GND smallest). CIN at pins 7/1, SW copper minimal, R4/R5 at FB pin 4 away from SW: ground plane under the part.

04 LITE HD SUPPLY SWITCH TPS22810



New block from TI SLVSDH0C (D15, D34; 53 BR--10/22/24).
P4-2 / BR--10: U5 pin on +5V_BST, ahead of the mux: USB never reaches the O4. C17 1 uF CIN on +5V_BST (one cap with U3 VIN2 if P5 brings the pins within -2 mm). C16 CT 10 nF: -4.7 V/mv, 0.2 A inrush per 4 uF. QOD tied to YOUT: the internal RP8 empties +5V_HD when off.
U5_EN (bom_plan O4_EN) holds exactly R9 33k / R10 18k from +BATT. R87 510k from +5V_HD. C129 100 nF, D2 anode and U5 pin 3 (netlist checked), D2 BAS16LD cathode + R11 2.4k pull-down on HD_EN (GPIO27, PINIO1).
P4-1 (VENR 1.35-1.30 V, VENF 1.08-1.18 V, +5V_HD 4.93-5.37 V): EN = 0.345 x cell (+0.0223 x +5V_HD when on). Nominal R: on at 3.27-3.77 V, shed at 2.78-3.10 V; 1 % R: on 3.27-3.77 V, shed at 2.78-3.10 V; 1 % R: on 3.27-3.77 V, shed at 2.78-3.10 V; 1 % R: on 3.27-3.77 V, shed at 2.78-3.10 V;
GPIO27 low or undriven: EN <= 0.715 V + 66 uA x 2.4k = 0.87 V < VENF 1.08 V, also with +5V_HD still up. GPIO27 high: D2 reverse >= 1.7 V; 1 uA hot leakage moves EN 11 mV (shed 0.03 V); R11 draws 1.4 mA.
Filter: 11.4 kOhm x 100 nF = 1.14 ms (140 Hz): the 96 kHz ESC ripple (0.21 V p-p at EN) falls to < 0.18 V. D2 + R11 discharge it below VENF in 0.25 ms; turn-on from the clamp -1.6 ms at 4.2 V. USB only (back-fed 2.5-2.8 V + 0.3 V ripple). EN <= 1.07 V < VENR. EN <= 1.62 V << 1.8 V abs max.

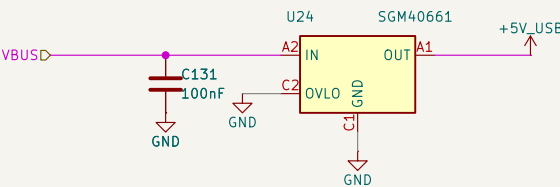
CAMERA 5 V FILTER



Source: OpenFC-Lite-Mini/hardware/power.kicad_sch (camera 5 V feed), with the commons 120 Ohm 0201 bead (blocks/LDO_IMU_TPS7A203S).

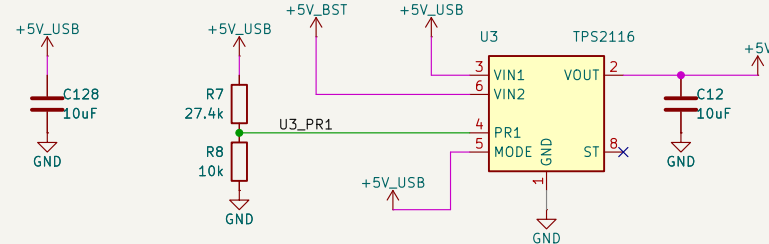
FB1 BLM03PX1225N1D: 120 Ohm at 100 MHz, 0.9 A, 160 mOhm: the 0.12 A camera drops 19 mV; it keeps the boost's 1 MHz and the camera's own switching apart. C15 10 uF X6S after the bead. +5V_CAM feeds the CAM pad J24 and the camera plug J32 (PAD5); PWR_FLAG because FB1 is passive. Feedback BLM03PX2205N1D (1.45 A).

USB INLET OVP SGM40661



New block from the SGMR0 SGM40659/60/61 dataset REVA (D57).
IN = YBUS (J31 pin 4, PADS). OUT = +5V_USB. OVLO (C2) to GND selects the internal threshold: 5.75–6.12 V rising, 5.45–5.66 V falling. IN 28 V abs max (40 V surge), 2.5–28 V range, UVLO 2.4 V. 15 ms debounce, then a 2 ms soft turn-on: a cable LC ring (simulated <= 10.8 μ s at IN) never reaches TPS2116 VIN1 (6 V abs max), and the soft turn-on is the YBUS surge limit USB asks for above 10 μ s downstream.
C131 100 nF X65 25 V at the IN balls (datasheet CIN 0.1 μ F; also takes ESD charge). COUT = C132 100 nF at U3 VIN1 (<= 100 μ F allowed). RON 27 / 45 mOhm: 11–19 mV at 0.42 A. TA < 40.85 $^{\circ}$ C.
Open: a non-compliant source at 5.5–6.12 V would pass before the OVLO trips (TPS2116 VIN1 5.5 V recommended, 6 V abs max); USB sources are <= 5.25 V (5.5 V with PD).

5 V POWER MUX TPS2116



New block from TI SLVSGF1A, priority mode (D24). C128 = the VIN1 cap: place it at U3 pin 3 (TI 9-10).
MODE tied to VIN1 (+5V_USB). PR1 = VIN1 x 10k / 3.74k: USB is selected above VREF x 3.74 = 3.44-4.04 V (VREF 0.92-1.08 V;
3.37-4.10 V x 1 X) +5V_BST (VIN2) otherwise. ST (open drain) unused.
VIN2 +5V_USB holds VIN2 at 0V until C11, C12, C13, C14, C15, C16, C17 (netlist checked). USB selected: VIN2 stays off
(truth table "VIN2" x) and both channel blocks sense current: no USB never feeds +5V_BST or the O4.
P4-3: C128 10 uF of X65 0402 at VIN1 (TI CIN 1 uF); VIN1/MODE/PR1 peaks are bounded by U24. Unplug with a cell: +5v follows
VBUS to the PR1 threshold, then the 8 us break-before-make gap: dv = 0.7 X 8 us / (C12 + C13 + C15, 3.4 uF each at 3.4
V). Murata curve = 0.55 V + +5V = 3.19 V typical; worst corner (3.44 V, -20 % caps, 2.75 V, +3V3 stays >= 3.07 V: the
C128 10 uF at VOUT; +5V hold-up). Its >= 7 uF of 0.1 uF caps (C15, C16) scopes it.
C128 10 uF at VOUT; +5V hold-up. Mux 40/60 Ohm; 0.04 V analog, 0.06 V at 1.48 A.

ESC3 EFM8 VDD BULK



Moved here from esc_cmn (extract sheet moves); that file is placed four times, so the ESC3--only part is drawn down POWER. Place C30 at U8 (ESC3 EFM8BB51) VDD pin; / GND, $\leq 3\text{ mm}$ (BR-05/BR-20 placement condition, P4-17). P4-19: 10 uF X6S 0402 (GRM155C80106M11D) = $2.4\text{--}3.9\text{ uF}$ effective at $4.35\text{--}3.0\text{ V}$ (Murata curve) against the EFM8's 1 uF. D69 makes the qvrr 5 uF bulk-MLCC distance a preference, so BR-20 stands (no 2,2 uF line); the critique confirms.

SHEET NOTES

POWER sheet, generated with hardware/tools/sch_build.py and sch_root.py (P4 power sheet); do not edit by hand.
R15 are power symbols (net = Value); +BATT_IN, +BATT, +5V_BST, +5V_USB, +5V, +5V_CAM, +5V_HD, +3V3, GND. Sheet pins: HD_EN (in),
CURR Sense_out (in), VBUS (in). Local nets: SHUNT_Sense_P(N/C), U2_SW, U2_FB, U3_PR1, U5_EN.
Closed here: P4-1, P4-2, P4-3, P4-7, P4-9, P4-12 (+3V3), P4-17 (BR-14 C13, BR-05-20 C30). P4-19; R11 is the GPIO27 pull-down P4-4
counts on; P4-13 is void (D69: C2, C5, C14 stay X5R, sized on DC bias). Values: research/DESIGN-SPEC.md 4.1, 4.2, 15.5; P4 hand calculations
reproduced in the hard notes.

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Sheet: /POWER/
File: power.kicad.

Title: Power: input, current sense, 5 V, 3.3 V, 04

Size: A3

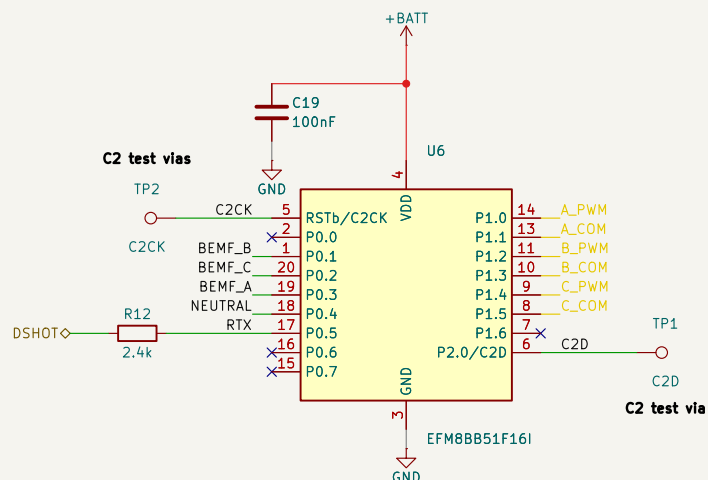
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

id: 2/14

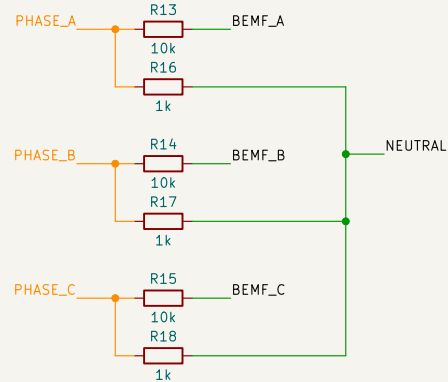
ESC MCU: EFM8BB51, Bluejay BB51 layout A



Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX; P1.0/P1.1 A_Fwm/A_Mux, P1.2/P1.3 B_P1.4/P1.5 C (PWM active high = N gate, OM active low = gate, DShot=pull); P2.0 DebugPin Pin 20 numbers: FEMBB51 data
VDD reg 1.0 abs max 6.1, P0.6, P0.7, P1.2, P2.0 DebbugPin connected (PINMAP 14).
VDD pin straight on +BAT+, no series (it must track the P sources); C19 10µF X65 at pin 4 /
GND pin 3. The data sheet 1 uF is the cell bulk C23 22 uF X65 (P4-17 / BR—05: ~ 3 mm from pin
4; the v3 cell puts it 3.4–3.9 mm away through In3, P5 measures, critique rules; ESC3 adds C30 10 uF
at its EFMB, drawn on POWER, P4–19). VDD abs max 5.5 V vs 4.35 V cell up & ripple + regen 4.7–5.25 V
(V3c).

DShot (P4-8): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD
+ 0.3 V abs max; R12 holds the clamp current to ~ 0.35 mA ($(3.33 - 2.5) \text{ V} / 2.4\text{k}$; about 0.1–0.2
mA after the clamp drop) against 50 mA per pin. RSTB/C2KCK: internal pull-up only (PINMAP F12).
C2D / C2CK: TP1 / TP2, mask-open filled test vias (D67), bench recovery only: NextPCB pre-programs
Bluejay before reflow, later updates go through Betaflight passthrough on DSHOT.

BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER_single_escsch (CERN-OHL v1.2, wire trace in resnet_08) 10k series from each phase to its comparator (BEMF, P0.4_V_Mux, BEMF_A, P0.1_V_Mux, P0.2_V_Mux). Each phase to the virtual neutral (P0.4_V_Mux, C_MPO negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to < 0.1 mA, 1k star, worst case one phase low and two high: 2/3 x 5.25 V / 1k = 3.5 mA, 12 mW per 0201 (50 mW part). P4=8: the netlist puts BEMF_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc_A_Mux 3 / B_Mux 1 / C_Mux 2 / V_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 C+6 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:

ESC1 M1 rear right: U6 C19 C23, R12~R18, Q29~Q31, TP1/TP2

ESC2 M2 front right: U7 C25 C29, R19~R25, Q32~Q34, TP3/TP4

ESC3 M3 rear left: U8 C31 C35, R26~R33, Q35~Q37, TP5/TP6 (+ C30 on POWER)

ESC4 M4 front left: U9 C37 C41, R33~R39, Q38~Q40, TP7/TP8

Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE_A/B/C (right, local label + in-sheet label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13).

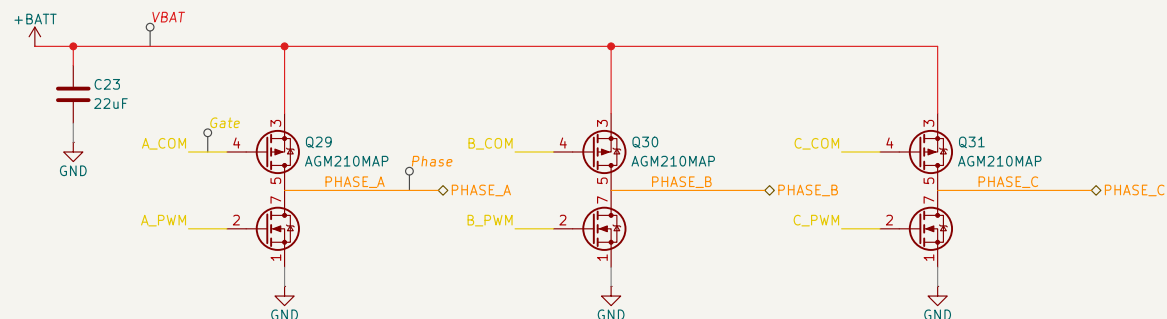
In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK, Netclass flags: Phase (PHASE_A), Gate (A.COM), VBAT (+BATT rail); setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.

Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFMB8B51 + 3x AGW210MAP (tinyPEPPER direct drive, Matrix 1S 51N1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP P412): batted first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, $I_{POR} \leq 390$ uA) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through $\approx 72k$ (pull-up + R12), i.e. ≤ 60 uA; 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix proxy (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot+unpowered (not in rev1).

Only if the PIO DIO divider is high, back-feeds +BATT through the EFM8 I/O divider (≈ 3.3 V), ≈ 2.5 uA, about 1 mA per channel; the EFM8s may cycle near POR, +5V, signals on V9 via the TPS2116 (10k fallback if V9 shows brown-out/cvtn),

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view: D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +12 V, VDS 20 V abs max vs VDD = 5.25 V regen peak. C23 22 uF X65 0603 is the ESC local bulk and the only commutation-loop capacitor (no leg caps, D56). Bluejay: stock BB51 A; the dead time is a V2 outcome (D30 reopened by D56; the Matrix II ships A_X_5_96 on this stage; the DS Qg of 23 / 33 nC discharges 3-5x with Ciss 950 pF, so V2 measures the gate edges).

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Sheet: /ESC1/

File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A

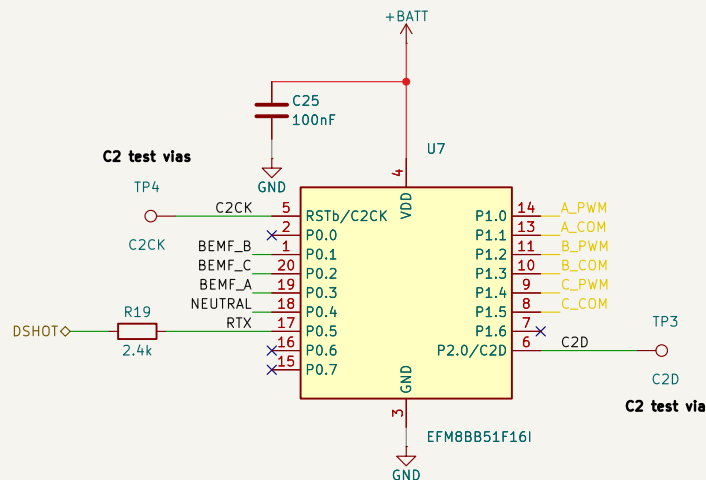
Date: 2026-10-09

KiCad E.D.A. 10.0.6

Rev: rev1

Id: 3/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



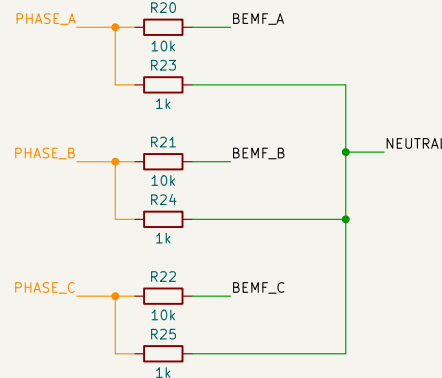
Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX; P1.0/P1.1 A_Pwm/A_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D; Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X65 at pin 4 / GND pin 3. The data sheet's 1 uF is the cell bulk C23 22 uF X65 (P4-17 / BR-05: <= 3 mm from pin 4; the v3 cell puts it 3.4-3.9 mm away through In3, P5 measures, critique rules; ESC3 adds C30 10 uF at its EFM8, drawn on POWER, P4-19); VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7-5.25 V (V3c).

DShot (P4-8): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to <= 0.35 mA ((3.33 - 2.5) V / 2.4k; about 0.1-0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP F12).

C2D / C2CK: TP1 / TP2, mask-open filled test vias (D67), bench recovery only; NextPCB pre-programs Bluejay before reflow, later updates go through Betaflight passthrough on DSHOT.

BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF_A P0.3, BEMF_B P0.1, BEMF_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to < 0.1 mA, 1k star, worst case one phase low and two high: 2/3 x 5.25 V / 1k = 3.5 mA, 12 mW per 0201 (50 mW part). P4-8: the netlist puts BEMF_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A_Mux 3 / B_Mux 1 / C_Mux 2 / V_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:

ESC1 M1 rear right: U6 C19 C23, R12-R18, Q29-Q31, TP1/TP2

ESC2 M2 front right: U7 C25 C29, R19-R25, Q32-Q34, TP3/TP4

ESC3 M3 rear left: U8 C31 C35, R26-R32, Q35-Q37, TP5/TP6 (+ C30 on POWER)

ESC4 M4 front left: U9 C37 C41, R33-R39, Q38-Q40, TP7/TP8

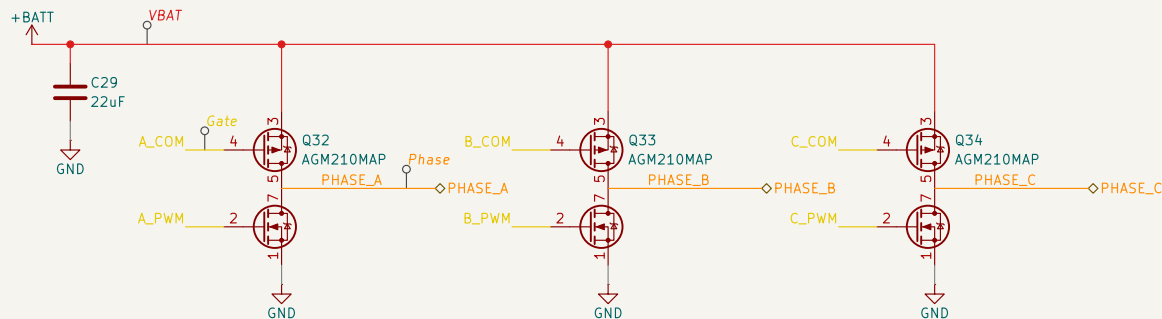
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail); setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.

Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA; 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode, <= (3.3 V - VF) / 2.4k, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, D5 VER2.73 p1 top view; D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +12 V, VDS 20 V abs max vs VDD <= 5.25 V regen peak. C23 22 uF X65 0603 is the ESC local bulk and the only commutation-loop capacitor (no leg caps, D56). Bluejay: stock BB51 A; the dead time is a V2 outcome (D30 reopened by D56; the Matrix II ships A_X_5_96 on this stage; the DS Qg of 23 / 33 nC disagrees 3-5x with Ciss 950 pF, so V2 measures the gate edges).

OpenDrone

Sheet: /ESC2/

File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4

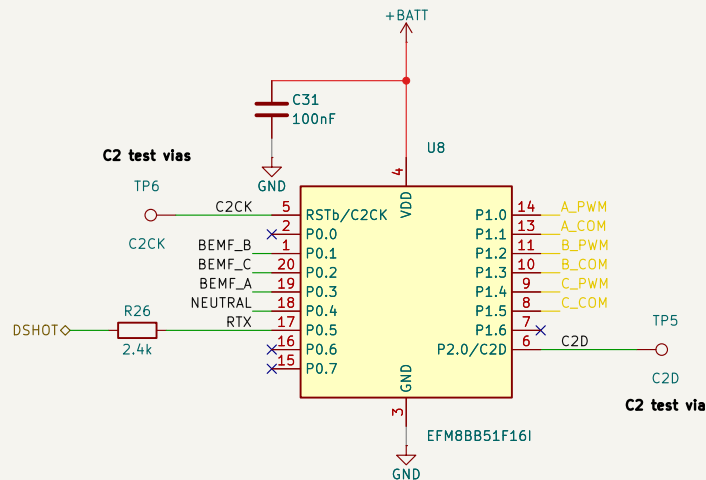
Date: 2026-10-09

Rev: rev1

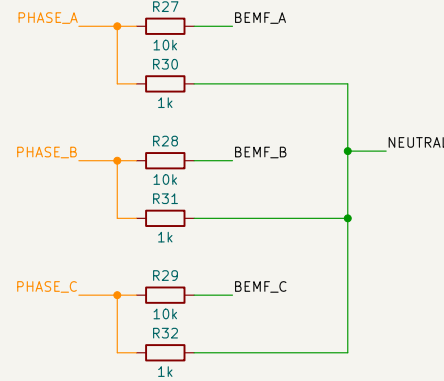
KiCad E.D.A. 10.0.6

Id: 4/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single_esc.sch (CERN-OHL v1.2, wire trace in research 08); 10k series from each phase to its comparator input (BEMF_A P0.3, BEMF_B P0.1, BEMF_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to < 0.1 mA, 1k star, worst case one phase low and two high: 2/3 x 5.25 V / 1k = 3.5 mA, 12 mW per 0201 (50 mW part). P4-8: the netlist puts BEMF_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A_Mux 3 / B_Mux 1 / C_Mux 2 / V_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:

ESC1 M1 rear right: U6 C19 C23, R12-R18, Q29-Q31, TP1/TP2

ESC2 M2 front right: U7 C25 C29, R19-R25, Q32-Q34, TP3/TP4

ESC3 M3 rear left: U8 C31 C35, R26-R32, Q35-Q37, TP5/TP6 (+ C30 on POWER)

ESC4 M4 front left: U9 C37 C41, R33-R39, Q38-Q40, TP7/TP8

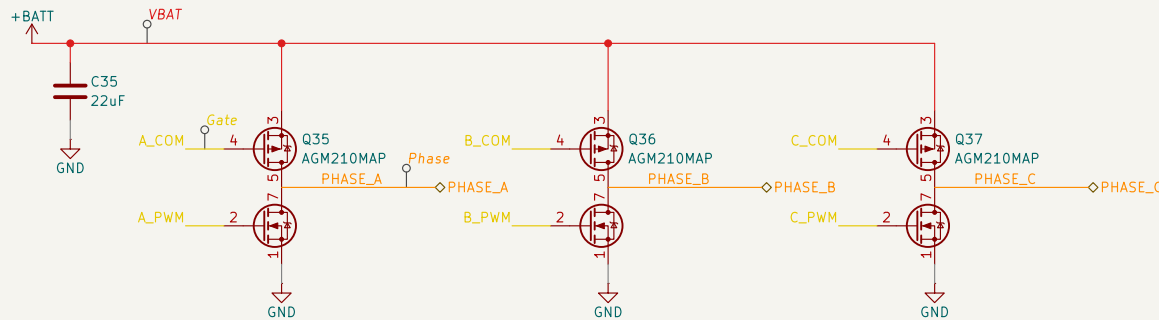
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail); setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.

Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA; 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode, <= (3.3 V - VF) / 2.4k, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, DS VER2.73 p1 top view; D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +12 V, VDS 20 V abs max vs VDD <= 5.25 V regen peak. C23 22 uF X65 0603 is the ESC local bulk and the only commutation-loop capacitor (no leg caps, D56). Bluejay: stock BB51 A; the dead time is a V2 outcome (D30 reopened by D56; the Matrix II ships A_X_5_96 on this stage; the DS Qg of 23 / 33 nC disagrees 3-5x with Ciss 950 pF, so V2 measures the gate edges).

OpenDrone

Sheet: /ESC3/

File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4

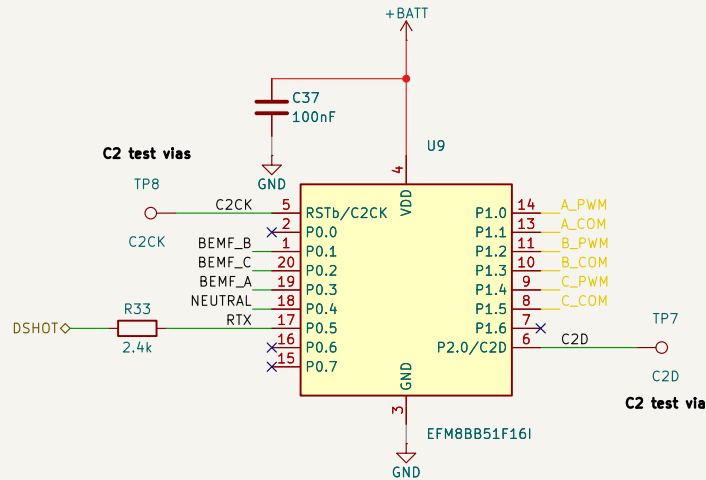
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 5/14

ESC MCU: EFM8BB51, Bluejay BB51 layout A



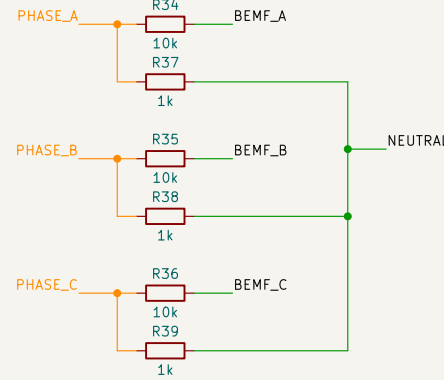
Pin use = Bluejay src/Layouts/BB51/A.inc (0368d11): P0.1 B_Mux, P0.2 C_Mux, P0.3 A_Mux, P0.4 V_Mux (neutral), P0.5 RTX; P1.0/P1.1 A_Pwm/A_Com, P1.2/P1.3 B, P1.4/P1.5 C (PWM active high = N gate, COM active low = P gate, all push-pull); P2.0 DebugPin = C2D; Pin numbers: EFM8BB51 data sheet Rev 1.0 Table 6.1. P0.0, P0.6, P0.7, P1.6 unused: no-connect (PINMAP 7.1).

VDD pin 4 straight on +BATT, no series R (it must track the P sources); C19 100 nF X65 at pin 4 / GND pin 3. The data sheet's 1 uF is the cell bulk C23 22 uF X65 (P4-17 / BR-05: <= 3 mm from pin 4; the v3 cell puts it 3.4-3.9 mm away through In3, P5 measures, critique rules; ESC3 adds C30 10 uF at its EFM8, drawn on POWER, P4-19); VDD abs max 5.5 V vs 4.35 V cell + ripple + regen 4.7-5.25 V (V3c).

DShot (P4-8): R12 2.4k at RTX. At a 2.5 V cell the 3.3 V DShot high sits up to 0.53 V over P0.5's VDD + 0.3 V abs max; R12 holds the clamp current to <= 0.35 mA ((3.33 - 2.5) V / 2.4k; about 0.1-0.2 mA after the clamp drop) against 50 mA per pin. RSTb/C2CK: internal pull-up only (PINMAP F12).

C2D / C2CK: TP1 / TP2, mask-open filled test vias (D67), bench recovery only; NextPCB pre-programs Bluejay before reflow, later updates go through Betaflight passthrough on DSHOT.

BEMF network: 10k series + 1k neutral star



Source: tinyPEPPER single_esc.sch (CERN-OHL v1.2, wire trace in research 08): 10k series from each phase to its comparator input (BEMF_A P0.3, BEMF_B P0.1, BEMF_C P0.2) and 1k from each phase to the virtual neutral (P0.4 V_Mux, CMP0 negative). No leg to GND: at 1S the phase stays within the cell + one body-diode drop, and the 10k limit the input clamp current to < 0.1 mA, 1k star, worst case one phase low and two high: 2/3 x 5.25 V / 1k = 3.5 mA, 12 mW per 0201 (50 mW part). P4-8: the netlist puts BEMF_A/B/C and NEUTRAL on U6 pins 19/1/20 and 18 = A.inc A_Mux 3 / B_Mux 1 / C_Mux 2 / V_Mux 4.

ESC channel, one file placed 4x. The references drawn are ESC1's; the root sheet offsets them per instance (ESC2 U+1 Q+3 R+7 TP+2, ESC3 x2, ESC4 x3), matching bom_plan.json:

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ESC2 M2 front right: U7 C25 C29, R19-R25, Q32-Q34, TP3/TP4

ESC3 M3 rear left: U8 C31 C35, R26-R32, Q35-Q37, TP5/TP6 (+ C30 on POWER)

ESC4 M4 front left: U9 C37 C41, R33-R39, Q38-Q40, TP7/TP8

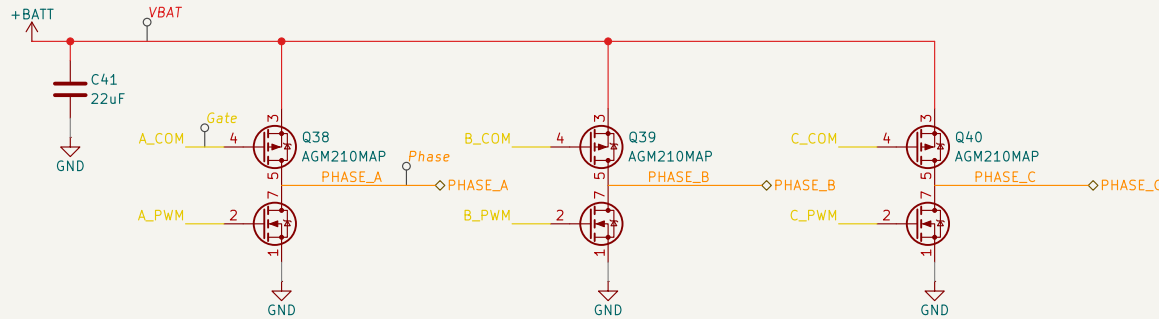
Pins: DSHOT (left, root net /MOTORn: FC GPIO25/24/23/22); PHASE_A/B/C (right, local label + hierarchical label, net /ESCn/PHASE_x, motor lands J4/J7/J10/J13). In-sheet nets: x_COM, x_PWM (Gate class), BEMF_x, NEUTRAL, RTX, C2D, C2CK. Netclass flags: Phase (PHASE_A), Gate (A_COM), VBAT (+BATT rail); setup_board.py patterns /ESC?/PHASE_?, /ESC?/?_COM, /ESC?/?_PWM assign the rest.

Source: copied from OpenAIO hardware/esc_channel.kicad_sch (frames, block titles, C2 test points at the MCU pins, three-column BEMF block, power stage along the bottom) and adapted: AT32F421 + NSG2065Q + 6x DOY180N03T become EFM8BB51 + 3x AGM210MAP (tinyPEPPER direct drive, Matrix 1S 5IN1 II stage, D56), the 3S divider becomes the tinyPEPPER 1S star.

Power-up (P4-11, PINMAP F16): battery first, the EFM8 runs from the cell within 0.4 ms (POR 1.76 V, tPOR <= 390 us) and the Bluejay bootloader turns on the 70-220k weak pull-ups (RTX included) while the RP2354A IOVDD still ramps (boost soft start + LDO, about 1-2 ms). Worst-case FC pad voltage in that window: the cell, up to 4.35 V (LiHV full), through >= 72k (pull-up + R12), i.e. <= 60 uA; 0.72 V over the 3.63 V unpowered-FT limit, current-limited. Accepted on Matrix parity (G473 FT pads about 4 V unpowered); V9 scopes the pad at power-up; fallback a 3.3 V-domain clamp or a DShot-path enable (not in rev1).

USB only: the PIO DShot idle-high back-feeds +BATT through the EFM8 I/O diode, <= (3.3 V - VF) / 2.4k, about 1 mA per channel; the EFM8s may cycle near POR, +5V stays on USB via the TPS2116 (10k fallback if V9 shows brown-out cycling).

Half-bridges A/B/C: AGM210MAP P+N, one package per phase



AGM210MAP (AGMSEMI, D5 VER2.73 p1 top view; D23 rule, P3 symbol/footprint check): 1 S1 N source = GND, 2 G1 N gate = x_PWM, 3 S2 P source = +BATT, 4 G2 P gate = x_COM, 5/6 + EP D2 (P drain) and 7/8 + EP D1 (N drain) = PHASE_x; the two drain pads are joined only by the phase island on the board (never split it). Gates straight from the EFM8: no gate resistors and no OR options (tinyPEPPER / Matrix II field record; bom_plan v3 has none); the about 0.15 A gate-edge current for tens of ns is a transient above the EFM8's 50 mA per-pin abs max, accepted as on the Matrix II (spec 4.4). VGS +12 V, VDS 20 V abs max vs VDD <= 5.25 V regen peak. C23 22 uF X65 0603 is the ESC local bulk and the only commutation-loop capacitor (no leg caps, D56). Bluejay: stock BB51 A; the dead time is a V2 outcome (D30 reopened by D56; the Matrix II ships A_X_5_96 on this stage; the DS Qg of 23 / 33 nC disagrees 3-5x with Ciss 950 pF, so V2 measures the gate edges).

OpenDrone

Sheet: /ESC4/

File: esc_channel.kicad_sch

Title: ESC channel: EFM8BB51 + 3x AGM210MAP

Size: A4

Date: 2026-10-09

KiCad E.D.A. 10.0.6

Rev: rev1

Id: 6/14

decoupling caps. place close to IC, one is enough when 2 pins are next to each other

vbat sense, 1/2 scale

current sense RC, 1.6 kHz

Crystal per datasheet. 10pF loading = 2x 15pF in series + -2.5pF stray

X1 YXC X252012MMB45-24 (2520, CL 10 pF, +/-0.01 ppm, -40..85 C) replaces the house
 TOSNINE part (-20..70 C); fallback: KYX K22120001210 (CL 12 pF, C42/C43 become 18 pF). XOUT
 1k series R40 (RPI guide, OpenFEC R5). Place X1, C42, C43, R40 at pins 21/22, short traces,
 nothing switching under them. Copied from OpenFEC-Lite-Mini rp2350a (crystal block).

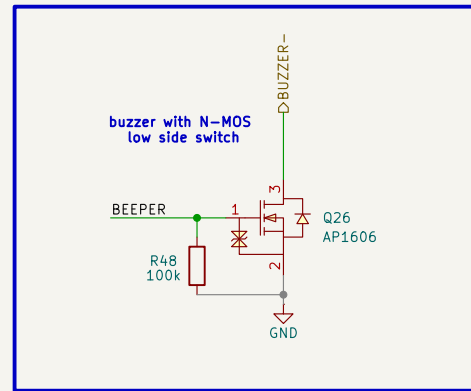
Boot strap + SWD test vias

R44 1k from QSPI_SS (pin 60, BOOT) to FC_BOOT: the shared boot button SW1 pulls FC_BOOT low through U8 (RX sheet, D65); low at reset = BOOTSEL / UF2. 1k, not the guide's 4.7k, so CSn <= 0.51 V with the diode drop (Vf 0.8 V). Keep the QSPI_SS-to-R44 stub <= 1 mm (in-package flash CS, RPI guide R6); any length after R44.

TP9 FCB, TP12 CLK (SWCLK), TP13 DIO (SWDIO): mask-open filled 0.35/0.20 vias (D67), no silk, out of the BOM and the CPL listed in the pinout (the house sheet leaves SWD open). Source: OpenFEC-Lite-Mini rp2350a BOOTSEL strap, button moved to the RX sheet.

[illegible]

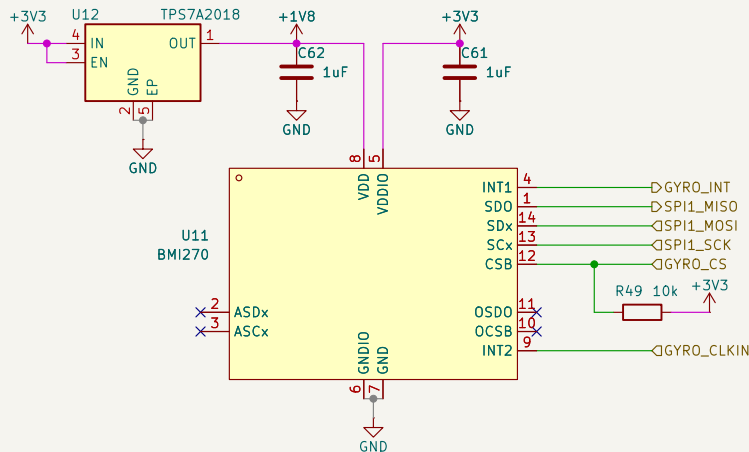
PIO2 framebuffer OSD: W, EN, SYNC consecutive



SHEET RP2350A, core type: OpenFC-Lite-*Mini/hardware/rp2350a.ikad.scf* (flow: *Review*), control block by layout with its layout, frames and text style. Changes: USB-C + CC removed (USB = J18 SJ31 + 07 on PADS; 27 series R42/R43 stay at U10, D52 BR=21 value); I2C0 pull-up removed (GPIO4/5 = UART1); the inverting WS2812 stage is not copied (F1, BR=04; GPIO8 drives the LED pad); status LEDs moved to the LED sheet; BOOTSEL button replaced by R44 + FC_BOOT (shared button on REX); SWD to test via (D67): VBAT divider 10k/10k; 12 MHz crystal swapped: VREG_AVDD 27r; two 4.7 uF on +V1 (C56 at VREG_FB, C134 at DVDD 23). GPIO map: *research/PINMAP.md* 2-3 (30 of 30 GPIOs, no benefit): *Beagleflight* config *PINMAP* 5.

[illegible]

Universal IMU wiring and LGA-14 land (BMI270, ICM-42688-P)



GYRO BMI270: U11, R49

Source: OpenFC-Lite-Mini imu.kicad_sch (universal LGA-14 land, SPI1 4-wire, R49), copied and adapted. BMI270 on the BOM; TDK ICM-42688-P qualified on the same land (V10 fly-off, one part per revision).

Pins 2/3 (BMI270 ASDx/ASCx, TDK RESV) and 10/11 (OCSB/OSDO, TDK RESV): no-connect. BMI270 DS Table 22: unused ASDx/ASCx go to VDDIO or stay open, never to GND (the house sheet grounded them for the LSM6DSV16X).

Pins 6/7 GND (TDK pin 7 RESV must be GND). VDDIO pin 5 = +3V3 (RP2354A I/O level). VDD pin 8 = +1V8; the BMI270 has no supply-order rule (DS 4.3). Abs max 4.0 V on VDD/VDDIO, I/O up to VDDIO + 0.3 V.

Pin 9: TDK CLKIN = GYRO_CLKIN (RP2354A GPIO6, PWM 32 kHz, driven only with the TDK part); BMI270 INT2, unused, left disabled. R49 10k; GPIO13 resets with its pull-down on, which would select the chip; R49 holds CSB >= 2.75 V.

P5: top side next to U12, LGA rotated 90 deg; GYRO_1_ALIGN from the placed orientation (PINMAP F11).

+1V8 GYRO LDO: U12, C61, C62 (D52 BR-09, P4-17)

Source: pcb-agent-commons blocks/LDO_IMU_TPS7A2033 (OpenFC-H7) topology with the 1.8 V part and no ferrite (fed from the +3V3 LDO): the house OpenFC-Lite-Mini also runs the gyro VDD from its own 1.8 V LDO. TPS7A2018 DQN: 1 OUT, 2 GND, 3 EN, 4 IN, 5 EP to GND. IN = EN = +3V3; VIN 1.6-6.0 V, VEN 0-6.0 V (abs max 6.5 V), EN never above IN. Load < 1 mA.

No 100 nF at the gyro: C61 = U12 CIN at BMI270 VDDIO pin 5, C62 = U12 COUT at VDD pin 8 (COUT >= 0.47 uF effective, ESR <= 100 mOhm; about 0.6-0.85 uF at 1.8 V). C61 counts in the +3V3 COUT sum (P4-12, POWER).

P5: U12 on the gyro side with OUT facing pin 8; C61 between U12 pin 4 and gyro pin 5, C62 <= 1.5 mm from pin 8 (P5-06), else re-place U12.

OpenDrone

Sheet: /IMU/

File: imu.kicad_sch

Title: Gyro BMI270 + 1.8 V LDO

Size: A5

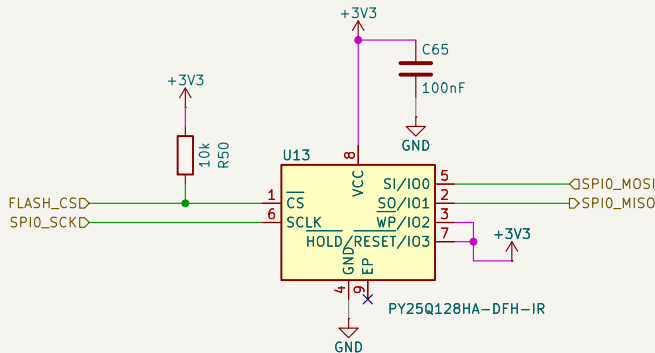
Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 8/14

SPI NOR PY25Q128HA: 16 MB blackbox on SPI0 (U13 C65 R50)



Placement (P5): beside the RP2354A on SPI0 (bom_plan region nor, top; either side works). U13, SPI0 and FLASH_CS stay ≥ 3 mm from the RX antenna hole (spec 4.9 and 11, DRU 'SPI0 3 mm from the RX antenna hole'): SPI0 harmonics fall in the ELRS band, V7 measures desense while logging. C65 at pin 8. Filled+capped 0.35/0.20 via-in-pad allowed in the lead lands, never in the tabs. RP2354A pads reset to 4 mA drive and slow slew (EMI lever without parts). Sourcing: no LCSC listing; NextPCB buys by MPN (D60), the quote confirms it. Same-pinout fallbacks in the Betaflight table (USON-8 4x4 land not staged): PY25Q128HA-QVH-IR, GD25Q128EQIGR.

Source: OpenFC-Lite-Mini hardware/blackbox.kicad_sch (rev2). Kept: layout, the 10k CS pull-up (R39 $\rightarrow$ R50) and the 100n supply cap (C43 $\rightarrow$ C65). Changed: its microSD (Card1) became this 16 MB SPI NOR (D66); the SPI0(SCK,MOSI,MISO) bus label became four hierarchical labels (sch_contract.json).

U13 Puya PY25Q128HA-DFH-IR: 128 Mbit, USON-8 3x3 (DF), 2.7-3.6 V, -40..85 C, fC 133 MHz, READ 03h 80 MHz, JEDEC ID 85 20 18 is in the Betaflight m25p16 table: USE_FLASH_M25P16, SPI0 PA18/PA19/PA20, FLASH_CS_PIN PA21 (PINMAP). SPI0 writes at 75 MHz and reads at 50 MHz: inside fC and fR.

WP# (3) and HOLD#/RESET# (7) tied to +3V3: the DS gives an internal pull-up only to the SOP-16 dedicated RESET#, so pin 7 must not float. Delivered SR = 00h, CR = 00h (QE 0, HOLD# function, WPS 0: no block lock) and Betaflight sends 1-bit SPI commands only, so IO2/IO3 never drive against the rail (the ERC pin_to_pin warning on pins 3/7 against a +3V3 power output is this deliberate tie).

R50 10k: CS# must follow VCC at power-up (DS 5.5). RP2354A pads reset with the pull-down on (PDE 1, RPD 36-113 kOhm at 3.3 V): CS >= 36/46 VCC = 0.78 VCC > VIH 0.7 VCC, so U13 stays deselected through reset and boot. U13 and the RP2354A IOVDD share +3V3: no pin above VCC + 0.5 V.

C65 100 nF X6S at pin 8, GND return straight to pin 4 (ICC max 13 mA read, 12 mA erase; adds 0.1 uF nominal to the +3V3 COUT sum of P4-12). EP (9) = two tie-bar tabs with no electrical function in the DS: no-connect, no vias in the tabs.

OpenDrone

Sheet: /BLACKBOX/

File: blackbox.kicad_sch

Title: Blackbox 16 MB SPI NOR PY25Q128HA

Size: A5

Date: 2026-10-09

Rev: rev1

KiCad E.D.A. 10.0.6

Id: 9/14

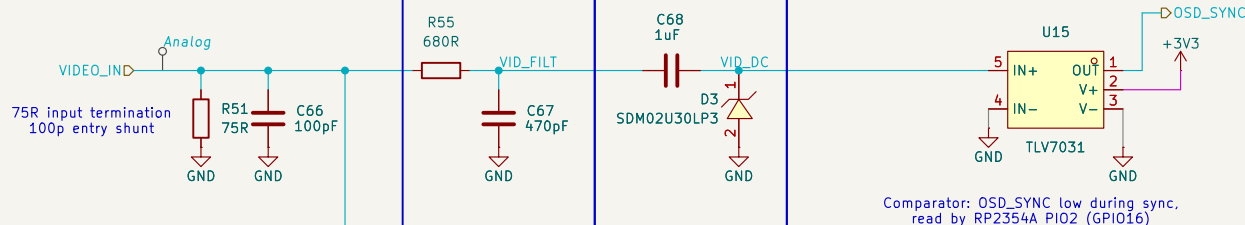
VIDEO PATH

Chroma Filter
fc = 500 kHz

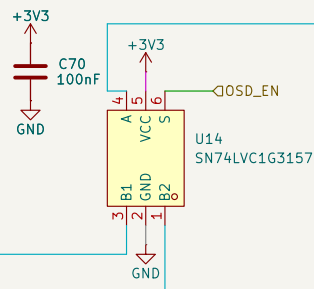
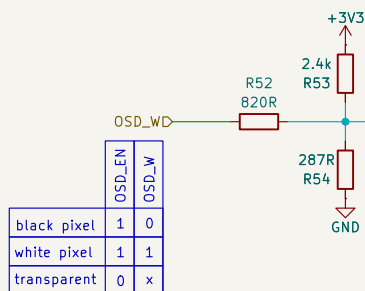
DC Restoration

Sync detection, VID_DC sync tips
clamped to $-V_F$ (about -0.1V to -0.2V)

U15 V+ (2) is decoupled by C70 in the switch cell,
one 100n shared with U14 (BR-13)



Resistor divider lets the OSD_W GPIO select
white (high: 1.05 V) or black (low: 0.27 V)



S low (OSD_EN low / reset): B1 camera
S high: B2 OSD level

Analog switch

VIDEO_OUT: the switch output drives the RTC6705 video
network on the VTX sheet directly (R58 1.2k || C104 33p,
R59 270R, C105 4.7u, R60 1.2k: OpenOSD-X values); the
house COS8051 buffer (U1, R46-R48, C41) is deleted (spec
4.8).
P4-10 levels at OSD_LVL (+3V3 3.30 V, OSD_W push-pull):
R53 || R54 = 0.352 V / 256R; with R52: black 0.268 V,
white 1.054 V, source 195R + switch ron 7-18R (house
0.261 / 1.064 V, 949R into the buffer). C105 AC-couples:
no DC load, the DC levels hold. AC load 2.0-2.7k (l. R60 to
GND): OSD swing $x0.90-0.93$, camera (37.5R) $x0.97-0.98$,
so OSD contrast 5-7 % under the camera's (V9). Pole 213R
 $x \sim 30\text{ pF} = 25\text{ MHz}$.
OSD_W: 2.74 mA high ($\leq 3\text{ mA}$, 2.79 mA at 3.36 V), 0.33
mA low.

Source: OpenFC-Lite-Mini hardware/osd.kicad_sch (rev2, flown), copied and
adapted: cells, texts, truth table and part spots kept (A3 -> A4). Same topology:
75R termination, 680R/470p chroma filter, 1u + SDM02U30LP3 sync-tip clamp,
TLV7031 comparator, SN74LVC1G3157 camera/level switch (contract source, spec
4.8).

Changed: COS8051 buffer deleted (output = VIDEO_OUT); divider R42 3.9k / R43
12k / R44 1.4k -> R52 820R / R53 2.4k / R54 287R (about 5x down, P4-10);
C39 + C42 -> one shared C70 100n X65 (D52 BR-13 + BR-18); C66 100p entry
shunt added (spec 11); refs per bom_plan (R41>R51, C38>C67, C40>C68, D9>D3,
U2>U15, R45>R55, R42-44>R52-54, U18>U14).
Symbols from the OpenDrone catalogue with TI pin numbers: TLV7031 X2SON 1 OUT,
2 V+, 3 V- (centre pad), 4 IN-, 5 IN+. SN74LVC1G3157 DTB 1 B2, 2 GND, 3 B1, 4
A, 5 VCC, 6 S. The house lib symbols used the EasyEDA pad order (consistent with
its own footprints), so the drawing is re-routed with the same nets: IN+ = VID_DC,
IN- = GND, B1 = camera, B2 = OSD level. The house sheet's rotated ground is
upright here.

Operation: OSD_EN low (RP2354A reset pull-down, OSD idle) -> B1, the camera
passes through. OSD_EN high -> B2: OSD_W high = white, low = black. No OSD
without camera sync (osd_pico.c overlays on detected sync). OSD_W/EN/SYNC =
GPIO14/15/16, consecutive, PIO2 (PINMAP).

Sync threshold (P4-10): 680R/470p = 498 kHz: colour burst $x0.11-0.14$
($+17-21\text{ mV}$), 4.7 us sync kept (tau 0.32 us). C68 + D3 (cathode on VID_DC)
clamp the sync tips to $-V_F$; the leakage-limited clamp current (uA average, tens
of uA at the tip) gives V_F about 0.1-0.2 V over temperature (l: DS 0.37 V max at
10 mA / 25 C, 0.20 V at 10 mA / 125 C). IN- at GND puts the trip point V_F
above the tip, 33-67 % of the 0.3 V sync, $\geq 0.08\text{ V}$ from tip and blanking after
VIO +8 mV and VHYS 2-17 mV. OSD_SYNC is low during sync (osd_tx.pio waits
for a low sync pin). IN+ at $-V_F$ stays inside abs max ($V_{EE} - 0.3\text{ V}$), at most 0.1 V
under the ROC input floor ($V_{EE} - 0.1\text{ V}$), no phase reversal (DS 6.4.1). tpd 3 us on
both edges (equal shift, as on the house board).

Ratings: +3V3 only (U14 1.65-5.5 V, U15 1.6-6.5 V). S: VIH 0.7 VCC = 2.31 V,
edges $\leq 10\text{ ns/V}$ (fast slew pads). AC-coupled camera sync tips (-0.3 to -0.4
V) are below the switch ROC (0.VCC) but inside abs max (-0.5 V , 50 mA; 37.5R
source), as on the house board: V9 checks. Supply: U15 0.9 uA, U14 $\leq 10\text{ uA}$,
divider 0.94-1.26 mA + OSD_W 2.74 mA. C68 X5R carries $\leq 0.6\text{ V DC}$: no
DC-bias loss (P4-13 hot-zone rule void, D69).

Placement (P5; P4-17 sheet note): bottom, region osd. BR-13: C70 is the only supply cap of U14 (VCC 5) and U15 (V+ 2): place it between the two
parts (2.5 mm apart). GND via-In-pad; the P5 check lists its distance to both pins. U15 centre pad (3, V-) to GND through a via (TI 7.4). R51 and
C66 at the video entry on the CAM pad J23 / plug J32 line: camera return into L2 at the CAM GND pad: video on L1/L6 over a plane with GND both
sides, never L3 (spec 11). R55, C67, C68, D3 next to U15 IN+. Video parts away from the boost and its inductor (preference, D69; v2 floorplan
3.3-8.6 mm; V5 video test).

Netclass Analog: VIDEO_IN (flag here), VIDEO_OUT, VID_FILT, VID_DC, OSD_LVL, OSD_SYNC by the setup_board.py patterns: OSD_W, OSD_EN Default.
Sourcing: exact MPNs from bom_plan: LCSC empty where bom_plan has none (C66, R52, R54; open item A9-LCSC). PS-01 (R54 -> 270R
RC0201FR-07270RL, shared with R59, -1 line): black 0.258 V, white 1.011 V, 187R, OSD_W 2.79 mA, also passes P4-10; proposed in contract-issues,
drawn = bom_plan.

OpenDrone

Sheet: /OSD/
File: osd.kicad_sch

Title: PIO analog OSD front end

Size: A4 Date: 2026-10-09

KiCad E.D.A. 10.0.6

Rev: rev1

Id: 10/14

SX1281 radio

2.4 GHz LPF + wire antenna

RFIO (22) -> RF_RFIO -> FL1
TDK DEA102700LT-6307A2 LPF
(fallback Johanson
2450FM07D0034T, re-match)
-> RF_RX_ANT -> AE1:

Boot / bind button

RX SHEET (P4 shard C). Sources: the SX1281 radio, 52 MHz TCXO, RGB LED and Wi-Fi pi match are COPIED from the house sheet OpenRX-Lite-UFL hardware/esp32c3_sx1281_lite.kicad_sch (same as OpenAIO rx_esp32c3_sx1281.kicad_sch): same symbol positions and wires, adapted to bom_plan refs/values/MPNs and to D52/D53/D58 (LDO mode, pin 5 to GND, UFL replaced by the wire-antenna hole, DEA102700LT LPF, 2450AT07AD0100001T chip antenna, grounds turned down). The ESP32-PICO-V3 block is new, from the Espressif ESP32-PICO series datasheet v1.3 Fig. 11 (D52 BR-01, D55); the shared boot / bind button is new (D65, Matrix II style). Pin plan: research/PINMAP.md 6. Parts: hardware/bom_plan.json. Radio and LED +3V3 come from the POWER sheet (U4 LP5912-3.3).

BOOT / BIND BUTTON (D65; new). SW1 Alps SKUBAAE010 (2.4 x 1.4 mm) from BOOT_SW to GND. D8 Diodes SDM02M30CLP3 dual Schottky, common cathode on BOOT_SW: A1 = FC_BOOT (RP2354A QSPI_SS through R44 1k, RP2350A sheet), A2 = RX_BOOT (ESP32 GPIO0, pin 23). Released: each strap sees only diode leakage ($I_R \leq 0.15 \mu A$ at 10 V, 25 °C), so GPIO0 keeps its internal pull-up and QSPI_SS its own level. Held at power-up: FC BOOTSEL + ESP32 download mode ($C_{SN} \leq 0.51 V$ vs VIL 0.8 V); GPIO0 $\leq 0.42 V$ vs VIL 0.825 V) = production and recovery flow (D67; replaces the RXB pad). Pressed while running: ELRS button (bind 1.5 s, Wi-Fi 5 s, reboot 12 s); 2.9 mA through R44, CS stays $\geq 2.62 V$, the FC keeps running. Bench V-btn: if GPIO0 dips below 2.5 V under XIP load while pressed, add 100 pF 0201 on BOOT_SW.

ESP32-PICO-V3 (U16), Espressif PICO datasheet v1.3 Fig. 11 / Table 4 (new block).
SUPPLY: VDDA (1, 43, 46), VDDA3P3 (3, 4), VDD3P3_RTC (19), VDD3P3_CPU (37) on +3V3; C82 10 uF + C76 100 nF at VDDA3P3 pins 3/4 (Fig. 11 C1/C2). The 4 MB flash, 40 MHz crystal, CAP1/CAP2 network and the VDD_SDIO capacitor are inside the SIP (Fig. 8); VDD_SDIO (26) stays open.
EN: CHIP_PU (9) = RX_EN, R56 10k to +3V3 + C73 1 uF (tau 10 ms), never floating.
STRAPS at reset (P4-6, checked on the netlist):
GPIO0 (23) RX_BOOT: internal 45k pull-up, D8 blocks -> 1 = SPI boot (SW1 held -> download).
GPIO22 (22) PA_EN: internal 45k pull-down + LP5907 EN 1 MOhm pull-down (VTX sheet), no external part -> 0.
GPIO5 (34) VTX_SPL_CLK: internal pull-up, R81 1k into the RTC6705 CMOS input -> 1 (default timing).
GPIO12 / MTDI (18): no connect, internal pull-down -> VDD_SDIO 3.3 V for the in-package flash (VDD_SDIO eFuse burn optional).
GPIO15 / MTDO (21): no connect, internal pull-up -> boot log on UOTXD (FC ignores it).
NOT CONNECTED: flash CMD/I011 (30) and CLK/I06 (31); NC pins 25, 35, 36, 44, 45, 47, 48; free I07/8/9/10 (32, 33, 28, 29: PSRAM on the -02 alternate), I020 (27), I35 (11), I38 (7), I39 (8).

SX1281 (U18), copied from OpenRX-Lite-UFL; SX1281MLTRT (D52 BR-25), SX1280MLTRT drop-in alternate (ranging engine only).
 LDO MODE (Semtech DS Rev 3.2 5.1, Fig. 15-4): DCC_SW (14) open, the sibling's 15 uH L4 deleted (the ELRS layout has no radio_dcdc); VDD_IN (2) tied to DCC_FB (12) = VREG_RADIO with C87 470 uF + C88 10 nF; VR_PA (1) C86 10 nF; VBAT_IO (11) C91 10 nF; C89 100 nF; C90 10 nF; C92 1 uF + radio local (P4-12 lever P5-06); the house sheet's extra 10 nF / 10 uF deleter (BR-06).
 PIN 5 IS GND (pin table) and is grounded here; the sibling sheets left it open (only XTB stays open with a TCXO, DS 15.2). DIO2 / DIO3 open.
 52 MHz TCXO X3 (kept, D53): clipped sine AC-coupled by C84 1 nF into XTA (4), C85 100 nF at VCC, XTB (6) open.
 insulated lambda/4 wire (about 28-29 mm, trimmed in V7) in a plated D0.5 hole; AE1 is DNP and out of BOM / CPL (wire fitted by hand). RF class (directive + */RF_); board rule areas RF_RX_ANT / RF_RX_EXIT.

[illegible]

The diagram shows a complex PCB layout with several key sections:

- Power Section:** A +3V3 power source connected to a 100nF capacitor (C83) and a network of resistors (R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100).
- LED Section:** Two LEDs are shown: RX_LED (blue) and RGB WS2812 (red). The RX_LED is connected to a 3V3 supply through a resistor (R1) and a diode (D1). The RGB WS2812 is connected to a 3V3 supply through a resistor (R2) and a diode (D2).
- RF Section:** An RF_WIFI_LANT module is connected to an RF_WIFI module via a 1.2pF capacitor (C132). The RF_WIFI module is connected to an LNA_IN module via a 1.2pF capacitor (C133).
- Sensor Section:** A sensor module is connected to a 3V3 supply through a resistor (R3) and a diode (D3). The sensor module is connected to a 3V3 supply through a resistor (R4) and a diode (D4).
- Connectors:** Several connectors are shown, including U0RXD/I03, U0TXD/O3, U0TXD/O4, U0TXD/O5, SDO/I07, and SDO/I08.

Wi-Fi (D58): LNA_IN (2) -> RF_WIFI 50 Ohm
CPW <= 8 mm -> C133 shunt (DNP) / R88 OR
series (fitted) / C132 shunt (DNP) -> AE2

D4 XINGLIGHT XL-1010RGB-C-2812B (copied from OpenRX-Lite-UFL) on +3V3. ELRS led_rgb GPIO22, one LED (overlay ledix_rgb_vtx [0]).

P4-15 DECIDED: KEEP ON +3V3. Datasheet: VDD 3.5-5.5 V, DIN VIH >= 2.8 V, 5 mA per colour with a 0.6 V sink knee. On +3V3 the supply is 0.2 V under the table minimum (blue / green may run below 5 mA; Vf + knee is about 3.4-3.6 V); DIN comes from a 3.3 V CMOS output on the same rail. On +5V the supply would be in range but the data guarantee is short (ESP32 VDD > 2.8 V, VDD = 2.64 V at a 400 mA impedance load, DS Table 18, vs VIH 2.8 V). Neither option sits fully inside the tables: the house RX circuit (OpenRX-Lite = UFL, OpenAoD) runs this LED on 3.3 V, so it stays. Bench V-LED on the first boards: red, green, blue and white at 3.3 V, cold and hot. Feedback: VDD pin to +5V (one trace; C83 is 25 V).

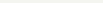
Johanson 2450AT07A01000011 (FEED 1/4, GND 2/3). Same pin as Espressif Fig. 11 (C4/L1/C3) and OpenRX-Lite-UFL (C22/L2/C21). The PiCO matches LNA_IN to 50 Ohm inside, so the network only tunes the antenna: values from V7 tuning (VNA at the feed or Johanson's service): R88 may become a Murata LQP03TN 0201 inductor. The A62 footprint carries the 5.0 x 3.2 mm keep-out (DRU 024).

wifi-antenna-edge-strip).

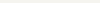
Pin	Signal	Function
17	DPA_BIAS_PWM	MTCK/IO13
18	VTX_SPL_DATA	MTMS/IO14
19	VTX_SPL_CLK	MTDO/IO15
20	VTX_SPL_CS	IO19
21	VTX_PWR_CTL	IO20
22	RX_LED	IO21
23	RX_LED	IO22

The schematic shows the ESP32-PICO-V3 module (U16) with its pins connected to a 4.9V source and GND. The TX pin is connected to the 4.9V source, and the GND pin is connected to GND. The module is labeled U16 ESP32-PICO-V3.

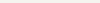
OpenDrone



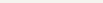


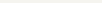


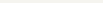


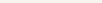


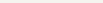




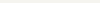




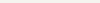


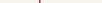


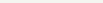


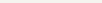


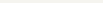


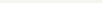


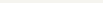












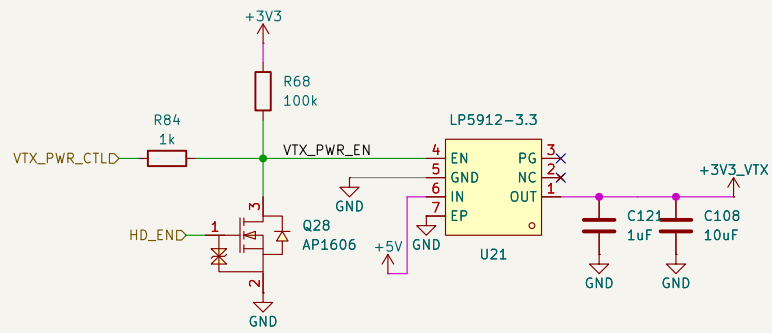
Sheet: /RX/
File: rx_esp32_sx1281.kicad_sch

Title: ELRS 2.4 GHz RX: ESP32-PICO-V3 + SX1281

Size: A3	Date: 2026-10-09	Rev: rev1
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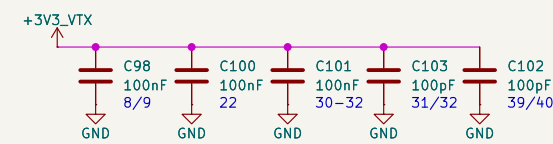
KiCad E.D.A. 10.0.6	Id: 11/14
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+3V3_VTX LDO + HD GATE

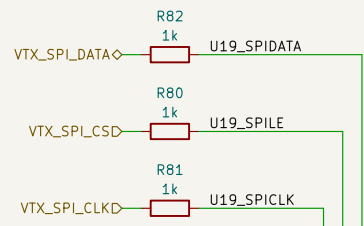


RTC6705 + 8 MHz CRYSTAL

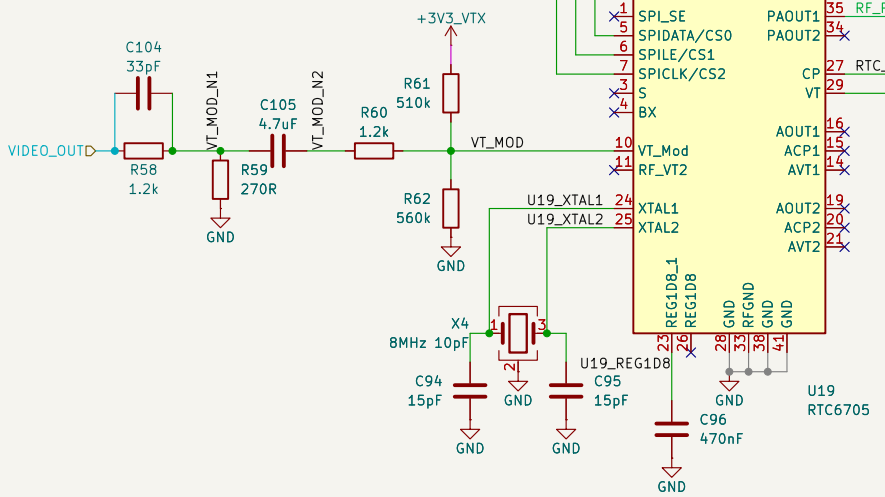
Decoupling at the RTC6705 supply pins (pin numbers next to each capacitor):



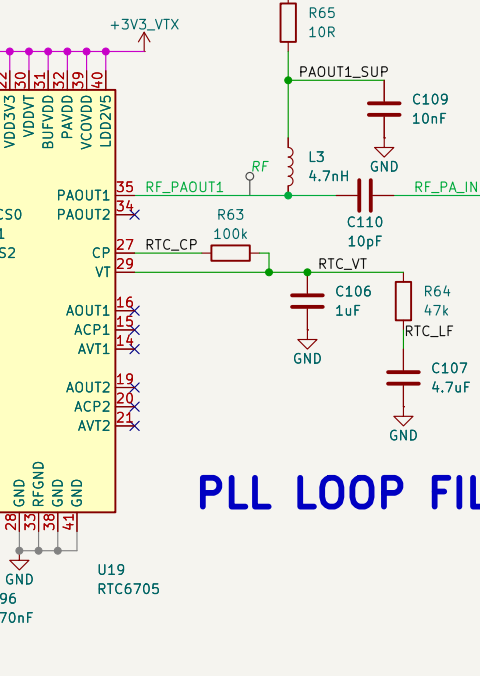
VTX SPI 1k SERIES



VIDEO MODULATION NETWORK

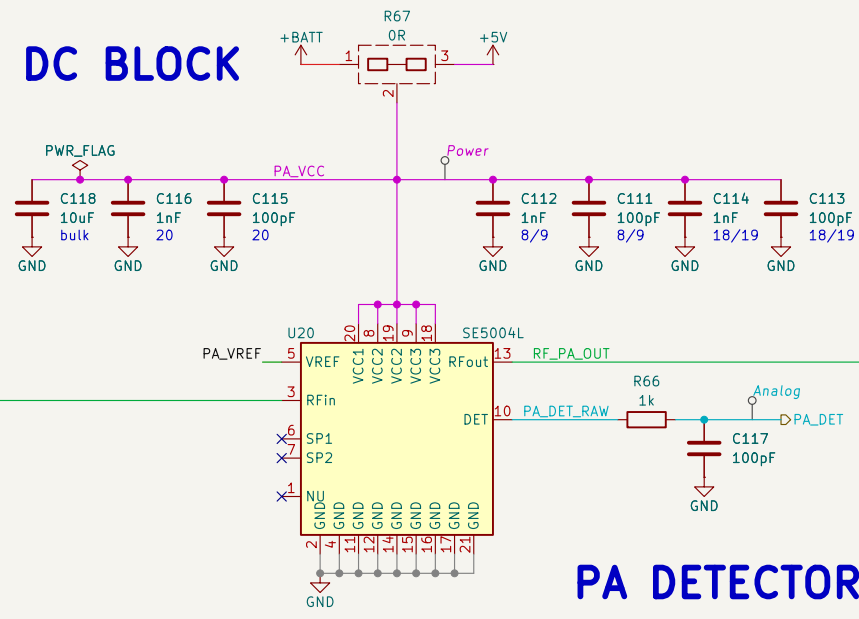


PLL LOOP FILTER

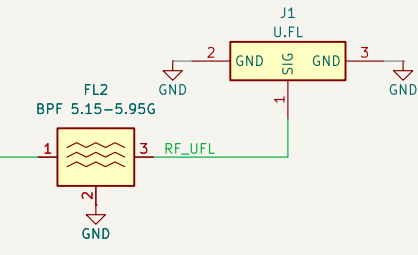


SE5004L PA + SUPPLY

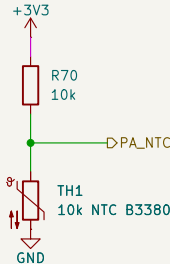
PAOUT1 FEED + DC BLOCK



HARMONIC BPF + U.FL



PA DETECTOR + NTC



SHEET SOURCE. Generated with hardware/tools/sch_build.py + sch_root.py (P4 shard script gen_vtx.py); no OpenDrone sibling sheet and no commons design block exists for an RTC6705 VTX (the gen_vtx.py clone has no hardware sheet), so the circuit follows the OpenOSD-X v1.01 reference schematic (RTC6705 supply pins, video network R24 C17 R25 C21 R29 R30 R31, loop filter R32 C18 R26 C19, PAOUT1 feed L4 C40 C43), the Skyworks SE5004L datasheet 2023938 and EK1 decoupling, DESIGN-SPEC 4.8 and decisions D52-D53 (BR-02, BR-03, BR-12, BR-15, BR-16, BR-17, BR-26), Refs, values, MPNs and footprints from hardware/bom_plan.json. Signal flow: ESP32 (RX sheet) SPI + PWM + EN on the left, RTC6705 in the middle, PA, BPF and U.FL on the right.

+3V3_VTX LDO + HD GATE (P4-4, P4-12, BOM-RED 7.1). Source: D12, D53 BR-03 (LP5912 and AP1606 are flown OpenDrone parts). U21 IN on +5V; its CIN is the shared 10 uF C13 on the POWER sheet between the U4 and U21 IN pins (BR-14; place within 2.4 mm of both). EN = VTX_PWR_EN; R68 100k to +3V3 (stock ELRS runs the VTX). R84 1k from ESP32 GPIO21 (patched-ELRS power-down hook, D54 OC-2; 1k per the 2026-10-09 ruling; GPIO21 low gives VEN = 0.03-0.04 V, guaranteed below VEN(OFF) 0.3 V). Q28 AP1606 drain (HD gate, D17/D53 BR-03). HD_EN high: AP1606 RDS(on) <= 220 mOhm at VGS 2.5 V sinks 3.3 mA (R84 1k with GPIO21 high, fine for the ESP32 pin; R68 adds 33 uA) -> VEN < 1 mV << VEN(OFF) 0.3 V. HD_EN low before Betaflight drives PINIO1: <= 0.2 V (BAS16LD clamp current into the 2.4k pull-down) vs AP1606 VGS(th) 0.4 V min (25 C); once PINIO1 is driven low the gate sits at about 0 V. COUT on +3V3_VTX: C108 10u + C121 1u + C122 1u + 3x 100n + 2x 100p = 12.3 uF nominal, 3.7-7.0 uF effective at 3.3 V with tolerance (typical DC-bias, I): inside LP5912 0.7-1.0 uF. PLACE C108 at the RTC6705 VCO pins 39/40 (VCO bulk, replaces the reference's 4.7 uF) and route +3V3_VTX C108 -> U22 directly (BR-03, P4-17), else C122 becomes C105A475MP5NRNFC.

PA REFERENCE (P4-4, P4-17). Source: D53 BR-03, TI SNWS798, U22 LP5907-2.85 from +3V3_VTX (BR-03); HD mode removes PA_VREF with the rail, EN = PA_EN straight from ESP32 GPIO2 (strap, 45k pull-down + internal 1M, no series R / no pull-down; BR-03, BR-12: VEN rated -0.3..6 V to GND). VREF 2.85 V +/-2 % (2.79-2.91 V vs SE5004L 2.80-2.90 V, V5 qualifies), C123 4.7 uF 0402 (BR-26, > 0.7 uF effective), C122 1 uF (about 0.5 uF effective) is enough only with the direct C108 -> U22 route, HD turn-off with VREF on: OUT may exceed VIN by one body-diode drop (<= 0.7 V, < 1 ms, <= 10 uJ) from C123) once U22 leaves dropout; V9 checks PA_VREF and +3V3_VTX < 0.5 V.

PNP PA DRIVE (P4-5, BR-02, BOM-RED 7.7). Source: D52 BR-02 (new stage, no sibling), GPIO13 10 kHz PWM -> 2x (1k, 1u) -> BC857BM base; emitter R73 Re to +3V3_VTX; collector Q27_C -> R65 10R -> PAOUT1_SUP -> L3 -> PAOUT1. Inverting, current mode, ngspice 42 + Nexperia BC857BM model (spice/pnp_spice.py, PAOUT1 sink model); Re 10R YOLO 32-59 mA over HFE 220-475 and -20..85 C, pinch-off >= 3248 counts, Q27 <= 114 mW (Ptot 130 mW at 85 C), base ripple 2.1 mV p-p, settles in 8 ms (ELRS 20 ms loop); GPIO13 floating at reset; 0 mA (45k pull-down case <= 19 mA, PA off: VREF low); HD mode: both junctions reverse-biased 3.3 V (VEBO 5 V), no back-power. Duty steps per 6 dB (P proportional to Idc squared) at Re 10R: 11-13 at I25 = 1 mA, 16-19 at 2 mA, 25-32 at 4 mA -> V8 measures the PAOUT1 current at 25 mA and V5 sets Re (27R gives >= 20 steps from 1.5 mA). Start value R73 = 10R (bom_plan).

VTX SPI (P4-4). Source: PINMAP F21, 1k series at the RTC6705 side of each line (place near the ESP32 per bom_plan; R80-R82 on the bottom at the RX MCU). ELRS HSPI 1 MHz, mode 0, LSB first, 25-bit frames sent as 32; tau = 1k x about 12 pF = 1.2 ns (26 ns 10-90 %S) against 11/12 -> 20 ns setup/hold inside a 500 ns half period; the three lines share the same RC (no skew), HD mode with stock ELRS wiring; <= 2.7 mA per high line into the unpowered RTC6705 clamps; CS idles high -> +3V3_VTX about 0.27 V on the LP5912 100 Ohm active discharge (V8 limit 0.5 V); bursts are rare (frequency changes only). SPI_DATA is bidirectional (3-wire reads through R82 on GPIO14, production test).

RTC6705 (P4-16, BOM-RED 7.5) checked pin by pin against OpenOSD-X v1.01: 23 REG1D8_1 -> C96 470n (ref C25 0.47u) = ref: 26 REG1D8 open (ref C27 DNI, BR-15) = ref: 17 AVDD_6.5 open (ref 3.3k + 1u to GND, no 3.3 V; BR-16 deviation, V5 checks no 6.5 MHz spur); 31/32 BUFVDD/PAVDD -> +3V3_VTX with C103 100p (ref C31 100p) + C101 100n shared with 3D; A0 LD02V5 -> +3V3_VTX direct with C102 100p (ref: 51R + 1u filter; lean-rule deviation V5 video SNR); 39 VCOVDD with C102 + the 10 uF C108 placed there (ref 4.7u). Pins 8/9 C98, 22 C100. Open: 1 SPL_SE (internal pull-up = SPI mode), 2-4, 11-16, 18-21 (no audio), 34, 36, 37, Xx exactly 8 MHz (ELRS R = 400). CL 10 pF; 15p + 15p in series + about 2.5 pF stray = 10 pF.

VIDEO + LOOP FILTER. Source: OpenOSD-X v1.01, Video: VIDEO_OUT (OSD switch, 75R termination on the OSD sheet) -> R58 1.2k || C104 33p -> R59 270R shunt -> C105 4.7u -> R60 1.2k -> VT_MOD. Bias R61 510k / R62 560k from +3V3_VTX (1.73 V) = OpenOSD-X R24 C17 R25 C21 R29 R30 R31 (R23 75R DNI there). Loop filter = OpenOSD-X; CF -> R63 100k -> VT_VT -> C106 1u and R64 47k + C107 4.7u to GND (R64 on the VT node as the reference R26; bom_plan nets synced at integration). Nets RTC_CP / RTC_VT / RTC_LF, VT_MOD_N1 / VT_MOD_N2 / VT_MOD. Keep CP/VT/LF copper short; >= 5 mm from the boost (M).

PAOUT1 FEED + DC BLOCK. Source: OpenOSD-X v1.01 L4 C40 C43 and R34 (fed by the BR-02 PNP instead of its NPN follower). PAOUT1 (pin 35, +2 dBm) -> L3 4.7 nH choke to PAOUT1_SUP (C109 10 nF X7R) <- R65 10R <- Q27_C; C110 10 pF DC block into RF_PA_IN (SE5004L pin 3 DC block required). Only L3, C109 and C110 sit in the 5.8 GHz part-free band between pin 35 and the PA (50 Ohm CPWG < 5 mm, 45 degree). Net class RF on RF_PAOUT1 / RF_PA_IN / RF_PA_OUT / RF_UFL (pattern V/RF-1, directive on RF_PAOUT1).

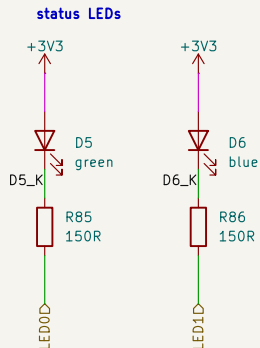
SE5004L PA (P4-18 closed; C119 deleted, D52 BR-17). Source: Skyworks SE5004L 2023938 and EK1. RfOut has an internal DC shunt to GND and only RFin is marked 'DC block required' (2023938 pin table); the BPF, U.FL and antenna carry no DC; a 10 pF 0201 at 5.8 GHz is past its about 3 GHz self-resonance (about +6 Ohm, |gamma| 0.06, 0.02 dB) so it is no match element either. RfOut -> FL2 directly, no DNP footprint on the line; P5 EM-checks the output match, PA_VCC = R67 selector: one 0402 0R from COM to +BATT (default) or +5V; VCC1/VCC2 <= VCC3 holds (one net). EK1 decoupling: In + 100p at 8/9, at 18/19 and at 20 on both package sides + C118 10 uF shared, no ferrite (spec 4.8). VREF from PA_VREF (abs max 3.6 V). SP1/SP2 (optional EVM caps) and NU open. EP + pins 2,4,11,12,14-17 to GND with a via field.

PA DETECTOR + NTC. Source: spec 4.8, DET (0.325 V no RF .. 1.0 V at 27 dBm) -> PA_DET_RAW -> R66 1k / C117 100p -> PA_DET (ESP32 GPIO4, ADC2) - the trace crosses sides, keep the RC at the ESP32 end of R66. TH1 MCP6X34H03F050R1 in the PA ring + R70 10k from +3V3 -> PA_NTC (GPIO34, ADC1, input only, no internal pull); 1.65 V at 25 C. BPF + U.FL: FL2 Walsin RFBPF1608060K9801C on the PA side away from the PA body (power handling at 0.71 W: 07); J1 U.FL with L2 cut out under the signal pad (RF_PAO_CUTOFF). >= 4 mm via at the tabs.

OPEN FOR THE CRITIQUE. (1) closed at integration; R84 = 1k (was 10k; VEN 0.30-0.33 V with GPIO21 low, at the LP5912 VEN(OFF) limit). (2) RTC6705 pin 11 RF_VT2 (audio subcarrier input) left open like the other audio pins: V5 spur check. (3) PA drive Re final value from V5/V8 (steps per 6 dB above). (4) LP5907 reverse excursion at HD turn-off (note above). V5.

OpenDrone

Sheet: /VTX/ File: vtx.kicad_sch		
Title: 5.8 GHz VTX: RTC6705 + SE5004L PA		
Size: A2	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6	Id: 12/14	



STATUS LEDs (D5 D6 R85 R86) source: OpenFC-Lite-Mini hardware/rp2350a.kicad_sch 'status LED' block (R52 + R53 2x 75R, D2 XL-1005UBC), copied and adapted: sink drive per hardware/sch_contract.json and research/PINMAP.md, 2x 75R -> one 150R 0201, second column for LED1.
 Circuit: +3V3 -> LED -> 150R -> GPIO, the pin sinks (contract). Betaflight default polarity, low = on, no LEDx_INVERTED: LED0_PIN PA7 (GPIO7, green D5), LED1_PIN PA26 (GPIO26, blue D6).
 Datasheet check (XINGLIGHT XL-1005UGC / UBC, 0402): VF 2.7-3.3 / 2.6-3.4 V at 20 mA, IF max 20 mA, PD 70 mW, VR 5 V. At a few mA the InGaN VF is about 0.2-0.3 V lower (inferred), so with VOL <= 0.5 V at 4 mA drive (RP2350 D5) the LED current is about 0.5-5 mA, typ. 2-3 mA, hard ceiling (3.3 - 2.4) / 150R = 6 mA: LED, 0201 resistor (<= 5.4 mW of 50 mW) and GPIO (4 mA setting, 100 mA IOVSS sum) all inside their ratings; brightness at 1-3 mA is about 17-135 mcd green and 4-36 mcd blue (lv scaled from 350-900 / 80-240 mcd at 20 mA), enough for an indicator. Reverse bias is 0 V when the pin drives high (IOVDD = +3V3).
 Reset state: RP2350 pads come up with the pull-down on (PDE = 1, 36-113 kOhm), so both LEDs may glow faintly (about 10-30 uA) until Betaflight drives the pins and in BOOTSEL mode. Accepted, cosmetic.
 Placement (P5): bottom side, fc_core region (bom_plan); the R may sit at either end of the LED, slow signal (D33). Pins LED0/LED1 = root nets /LED0, /LED1 (RP2350A).
 P4 closed here: bom_plan R85/R86 'P4 checks brightness vs VF' (above). P4-14 fields:
 RC0201FR-07150RL is a stocked Yageo RC_L part (Element14 4008725, RS 198-7568: 150R 1 % 50 mW 25 V); no LCSC number known, field left empty (contract open item A9-LCSC).

OpenDrone

Sheet: /LED/
 File: led.kicad_sch

Title: Status LEDs

Size: A5 Date: 2026-10-09
 KiCad E.D.A. 10.0.6

Rev: rev1
 Id: 13/14

Battery pads (1S)

The diagram illustrates the correct connection for a 1S battery. It shows two incorrect configurations crossed out with red 'X' marks and one correct configuration.

- Left (crossed out):** A battery symbol labeled J2 with a '+' sign is shown. The positive terminal is connected to a red dot, which is then connected to a terminal labeled +BATT_IN. The negative terminal is connected to a red dot, which is then connected to a terminal labeled GND.
- Middle (correct):** A battery symbol labeled J2 with a '+' sign is shown. The positive terminal is connected to a red dot, which is then connected to a terminal labeled +BATT_IN. The negative terminal is connected to a red dot, which is then connected to a terminal labeled GND.
- Right (crossed out):** A battery symbol labeled J3 with a '-' sign is shown. The positive terminal is connected to a red dot, which is then connected to a terminal labeled +BATT_IN. The negative terminal is connected to a red dot, which is then connected to a terminal labeled GND.

J2 '+' = +BATT_IN, J3 '-' = GND: BT2.0
Dip11 (22 AWG, user-fitted) through plated
D1.1 holes in 3.0 x 2.0 mm pads
(lib:BattPad_3.0x2.0mm_PTH1.1), >= 12 power
vias each into the pours (spec 4.10, D72
power plan: +BATT_IN into L4 + L1/L6, GND
into L2/L5). Silk '+' / '-' 2.0 mm and '1S'
(spec 13).

Copper only: DNP, out of BOM and CPL (LINEUP A16).

Motor lands M1–M4: plug-ready, Molex PicoBlade header DNP

The diagrams show the following connections:

- M1:** Phase to pin 2, 2 to pin 2, 1 to pin 1.
- M2:** 2 to pin 2, 2 to pin 2, 1 to pin 1.
- M3:** 2 to pin 2, 2 to pin 2, 1 to pin 1.
- M4:** 2 to pin 2, 2 to pin 2, 1 to pin 1.

All four diagrams are crossed out with a large red X, indicating that these connections are not supported or are deprecated.

Net names come from the ESC sheets (the root wires carry no label: /ESCn/PHASE_x). Netclass Phase: directive on M1_A here, the pattern /ESC?/PHASE_? assigns the rest. Phase copper runs from the AGM210MAP phase island straight into the land: solder area 0.30 mm and phase copper 0.20 mm from the edge (SOLDER_PAD class). M1 rear right (ESC2), M2 front right (ESC2), M3 rear left (ESC3), M4 front left (ESC4): Betafight quad-X.

USB: JST SH 1.0 mm 4-pin + ESD

VBUS is a hierarchical net (LINEUP A11) to POWER: U24 SGM40661 OVP IN + C131 100 nF take the hot-plug and over-voltage duty (D57). PWR_FLAG here: J31 is the only source of VBUS. A cable with pins 1/4 swapped puts -5 V on VBUS (unprotected, as on the Matrix).

User UART pads (PI0UART0)

Diagram illustrating the User UART pads (PI0UART0) and their connections:

- J16 (TP0) is crossed out with a red X. The connection is PI0UART0_TXD (green line).
- J17 (RP0) is crossed out with a red X. The connection is PI0UART0_RXD (green line).
- J18 (5V) is crossed out with a red X. The connection is +5V (purple line).
- J19 (GND) is crossed out with a red X. The connection is GND (blue line).

Silk TP0 RP0 5V GND; copper only (DNP, out of BOM and CPL, LINEUP A16).

LED strip + buzzer row

The diagram shows a 4-pin header row with the following connections:

- J33 GND:** Connected to GND (Ground).
- J20 LED:** Connected to LED_STRIP.
- J21 5V:** Connected to +5V.
- J22 BZ:** Connected to BUZZER.

Copper only (DNP, out of BOM and CPL, LINEUP A16).

Camera pads + JST SUR 0.8 mm plug

The diagram illustrates the correct wiring for the camera pads using a JST SUR 0.8 mm plug. It shows three incorrect connections that have been crossed out with red X's: J23 CAM, J25 GND, and J24 5V. A new connection, J32 CAM, is shown with a yellow background. This connection links the VIDEO_IN signal to the GND and +5V_CAM power lines.

HD pads (DJI O4 Lite)

The diagram shows four HD pads (J26, J27, J28, J29) on a PCB. Each pad is crossed out with a red X. J26 is labeled VHD and has a purple arrow pointing to +5V_HD. J27 is labeled GND and has a grey arrow pointing to GND. J28 is labeled TX1 and has a green arrow pointing to UART1_TXD. J29 is labeled RX1 and has a green arrow pointing to UART1_RXD.

Copper only (DNP, out of BOM and CPL, LINEUP A16).

Mounting holes



H1

MountingHole 3.5



H2

MountingHole 3.5



H3

MountingHole 3.5

H1–H3: the 25.5 mm M1.4 pattern (D73) is cut in Edge.Cuts (ear holes with flange keep-outs, spec 7), so the symbols are excluded from the board (FLOORPLAN 'For P4'), BOM and CPL; footprint kept as reference. No net (non-plated; GND annulus = 016), LOG01 (incutec, B.Silkscreen) is a board-only footprint, no symbol.

Wired on the bench: J31 pin 1 = GND (prototype 1), J32 pin order (gate G1), motor-header temperature (V4).

OpenDrone

Sheet: /PADS/
File: pads.kicad_sch
Title: Pads, motor lands, camera, HD, USB

Size: A3	Date: 2026-10-09	Rev: rev1
KiCad E.D.A. 10.0.6		Id: 14/14